

```
<!DOCTYPE html>
<html lang="hi">
<head>
  <meta charset="UTF-8" />
  <title>Gesture Controlled Home
Automation</title>
  <style>
    video, canvas {
      border: 2px solid black;
      width: 480px;
      height: 360px;
      display: block;
      margin: 10px auto;
    }
    #status {
      font-size: 24px;
      text-align: center;
      margin-top: 10px;
      font-weight: bold;
    }
  </style>
</head>
```

```
<body>
  <h1 style="text-align:center;">Gesture
Controlled Home Automation</h1>
  <video id="webcam" autoplay
playsinline></video>
  <canvas id="canvas"></canvas>
  <div id="status">डिवाइस का स्टेटस: OFF</div>

  <!-- Tensorflow and Handpose Models -->
  <script src="https://cdn.jsdelivr.net/npm/
@tensorflow/tfjs-core"></script>
  <script src="https://cdn.jsdelivr.net/npm/
@tensorflow/tfjs-converter"></script>
  <script src="https://cdn.jsdelivr.net/npm/
@tensorflow-models/handpose"></script>

  <script>
    const video =
document.getElementById('webcam');
    const canvas =
document.getElementById('canvas');
    const ctx = canvas.getContext('2d');
    const statusDiv =
```

```
document.getElementById('status');
  canvas.width = 480;
  canvas.height = 360;

let model;
let deviceOn = false;

async function setupCamera() {
  const stream = await
navigator.mediaDevices.getUserMedia({vid
eo:true});
  video.srcObject = stream;
  return new Promise(resolve => {
    video.onloadedmetadata = () => {
      resolve(video);
    };
  });
}
```

```
async function loadModel() {
  model = await handpose.load();
  console.log("Handpose model loaded.");
}
```

```
function isThumbsUp(landmarks) {  
  // Simple check: Thumb tip y should be  
less than index finger pip y (hand up)  
  return landmarks[4][1] < landmarks[6]  
[1];  
}
```

```
async function detectHands() {  
  const predictions = await  
model.estimateHands(video);  
  
ctx.clearRect(0,0,canvas.width,canvas.height);  
  
ctx.drawImage(video,0,0,canvas.width,canvas.height);  
  
  if(predictions.length > 0) {  
    const landmarks =  
predictions[0].landmarks;  
  
    // Draw red dots on hand landmarks
```

```
for(let i=0; i<landmarks.length; i++) {  
  const [x,y,z] = landmarks[i];  
  ctx.beginPath();  
  ctx.arc(x,y,5,0,2*Math.PI);  
  ctx.fillStyle = "red";  
  ctx.fill();  
}
```

```
if(isThumbsUp(landmarks)) {  
  if(!deviceOn) {  
    deviceOn = true;  
    statusDiv.textContent = "डिवाइस का  
स्टेटस: ON";  
    speak("लाइट चालू");  
  }  
} else {  
  if(deviceOn) {  
    deviceOn = false;  
    statusDiv.textContent = "डिवाइस का  
स्टेटस: OFF";  
    speak("लाइट बंद");  
  }  
}
```

```
}  
  requestAnimationFrame(detectHands);  
}
```

```
function speak(text) {  
  const msg = new  
SpeechSynthesisUtterance(text);  
  msg.lang = 'hi-IN';  
  speechSynthesis.speak(msg);  
}
```

```
async function main() {  
  await setupCamera();  
  await loadModel();  
  detectHands();  
}
```

```
  main();  
</script>  
</body>  
</html>
```

